



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

PAPER 1 Multiple Choice

6031/1

NOVEMBER 2019 SESSION

1 hour

Additional materials:

Data Booklet

Scientific/Electronic calculator

Multiple Choice answer sheet

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 15 printed pages and 1 blank page.

Copyright: Zimbabwe School Examinations Council, N2019.

QZIMSEC N2019

[Turn over

Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

- 1 What is the volume of oxygen gas produced, measured at s.t.p., when 2.5 A is passed through dilute sulphuric acid for 1 850 seconds using platinum electrodes?

A 268.40 cm³
B 536.79 cm³
C 1 073.00 cm³
D 1 149.60 cm³

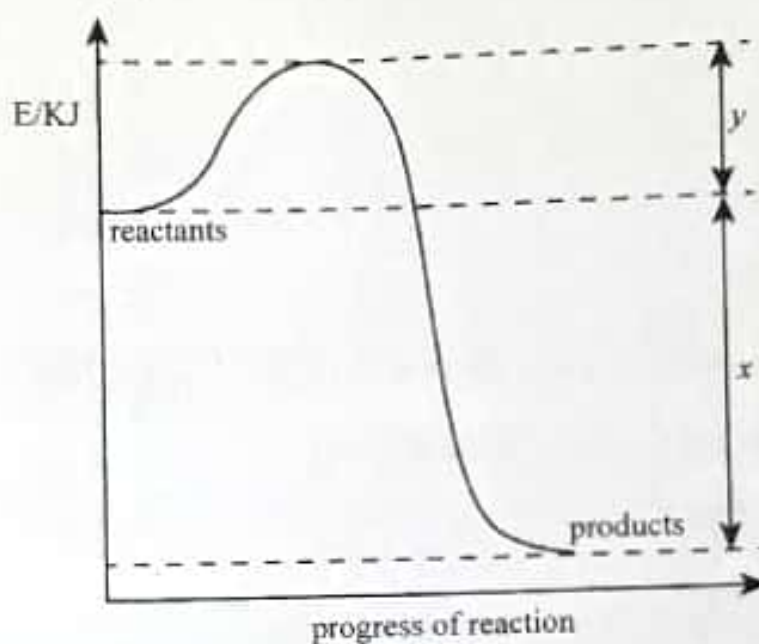
- 2 A volume of 15.00 cm³ of 0.040 mol dm⁻³ lead (II) nitrate was mixed with 15.00 cm³ of 0.040 mol dm⁻³ sodium chloride.

$$[K_{sp}: \text{PbCl}_{2(s)} = 1.7 \times 10^{-5}]$$

Which statement gives the correct result?

- A a clear solution with no precipitate is formed.
B PbCl₂ precipitates and excess Pb²⁺ ions will remain in solution
C PbCl₂ precipitates and excess Cl⁻ ions will remain in solution.
D PbCl₂ precipitates, and no excess ions will remain in solution.

- 3 An energy diagram for a reaction is shown:



The activation energy of the reverse reaction is given by the value of

- A x .
 B y .
 C $x + y$.
 D $x - y$.
- 4 A 0.10 mol dm^{-3} solution of a weak acid, HA, has a pH of 4.8.
 What is the value of pK_a ?
- A 3.8
 B 4.8
 C 8.6
 D 9.6
- 5 What is the percentage of ^{71}Ga given that its isotopes ^{69}Ga and ^{71}Ga have a relative isotopic mass of 69.8?
- A 0.4%
 B 0.6%
 C 40%
 D 60%

- 6 Flask A contains 2 dm^3 of helium at 4 kPa and flask B contains 4 dm^3 of neon at 2 kPa .
If the flasks are connected at constant temperature, the final pressure is

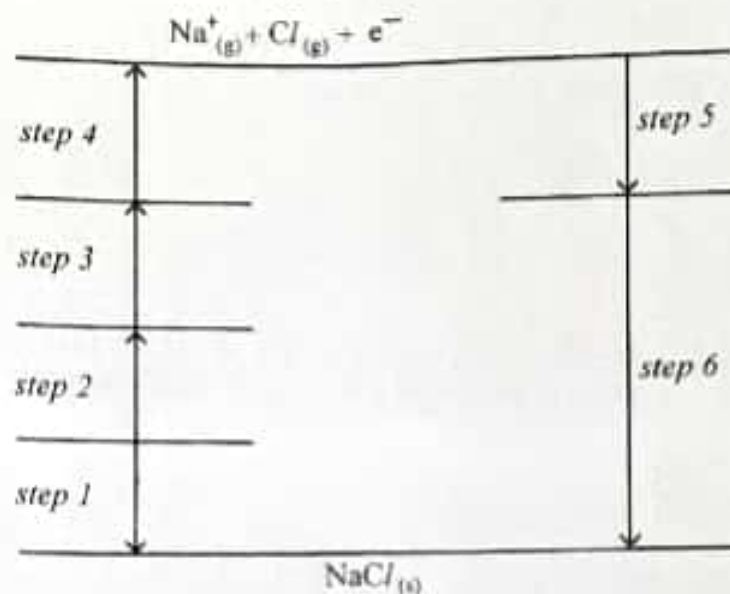
- A $\frac{2}{3} \text{ kPa}$
- B $\frac{4}{3} \text{ kPa}$
- C $\frac{5}{3} \text{ kPa}$
- D $\frac{8}{3} \text{ kPa}$

- 7 The given equilibrium exists in a gaseous system: $2X_{(aq)} \rightleftharpoons Y_{(aq)} + 2Z_{(aq)}$.

What are the units of K_c for the equilibrium?

- A $\text{mol}^2 \text{ dm}^{-6}$
- B $\text{mol}^{-2} \text{ dm}^6$
- C mol dm^{-3}
- D $\text{mol}^{-1} \text{ dm}^3$

- 8 The diagram shows the Born-Haber cycle of NaCl.



Which set correctly shows the chemical equations for steps 1 and 6?

	1	6
A	$2\text{Na}_{(s)} + \text{Cl}_{(g)} \rightarrow 2\text{NaCl}_{(s)}$	$2\text{Na}_{(g)} + 2\text{Cl}_{(g)} \rightarrow 2\text{NaCl}_{(s)}$
B	$\text{Na}_{(s)} + \frac{1}{2}\text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$	$\text{Na}_{(g)} + \text{Cl}_{(aq)} \rightarrow \text{NaCl}_{(s)}$
C	$\text{Na}_{(s)} + \frac{1}{2}\text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$	$\text{Na}_{(g)} + \text{Cl}_{(g)} \rightarrow \text{NaCl}_{(s)}$
D	$\text{Na}_{(g)} + \frac{1}{2}\text{Cl}_{2(g)} \rightarrow 2\text{NaCl}$	$\text{NaCl}_{(s)} \rightarrow \text{Cl}_{(aq)} + \frac{1}{2}\text{Cl}_{2(g)}$

- 9 A mass of 1.00 g of NH_4Cl was dissolved in 30.00 cm^3 of $3.00 \text{ mol dm}^{-3} \text{ NH}_3$.

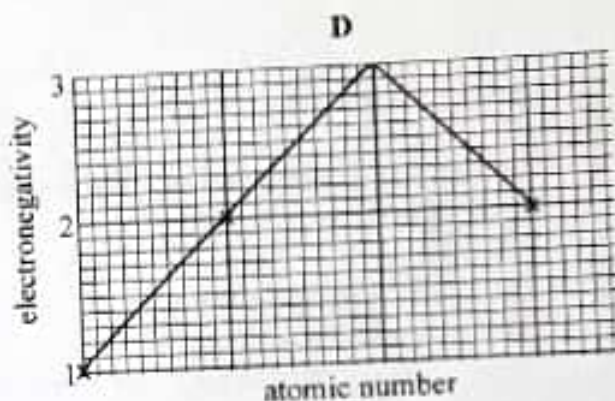
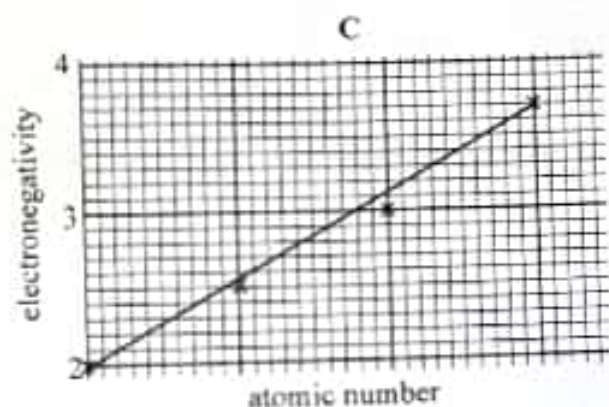
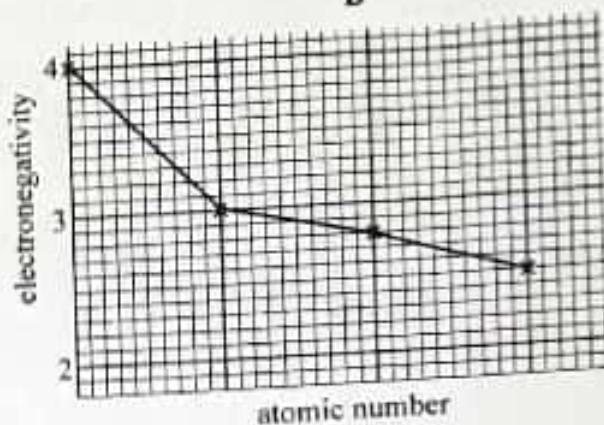
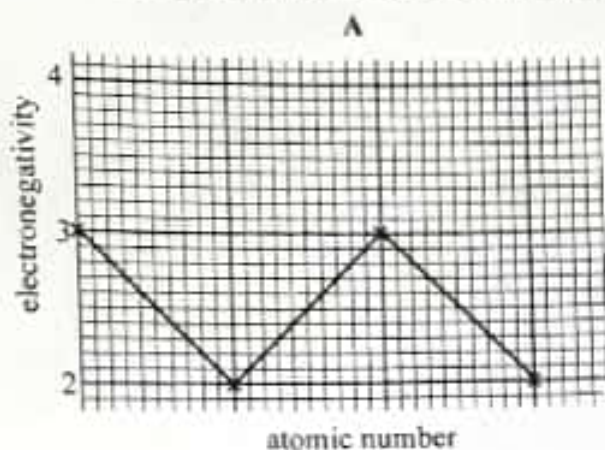
$$[K_b: (\text{NH}_3) = 1.8 \times 10^{-5}]$$

What is the $[\text{H}_3\text{O}^+]$ in the solution formed?

- A $5.5 \times 10^{-10} \text{ mol dm}^{-3}$
 B $2.7 \times 10^{-9} \text{ mol dm}^{-3}$
 C $1.2 \times 10^{-10} \text{ mol dm}^{-3}$
 D $1.4 \times 10^{-12} \text{ mol dm}^{-3}$

- 10 An electric current was passed through molten calcium chloride, producing 2.0 g of calcium metal at the cathode.
- What mass of chlorine was produced at the anode?
- A 1.78 g
 - B 3.55 g
 - C 4.00 g
 - D 7.10 g
- 11 Which two substances will remain, when a sample of an alloy of tin and lead is heated in excess oxygen and allowed to cool at room temperature?
- A Sn and PbO
 - B SnO and Pb
 - C SnO and PbO₂
 - D SnO₂ and PbO
- 12 Aluminium oxide is dissolved in molten cryolite during the extraction of aluminium by electrolysis because the cryolite
- A provides the ions needed to carry the current.
 - B reduces the high melting point of the electrolyte.
 - C reacts with the aluminium oxide to form ions.
 - D removes impurities in the aluminium oxide.
- 13 The compound PCl₅ is known, but no chemist has been able to make NCl₅ because
- A the covalent bond in the nitrogen molecule is too strong.
 - B nitrogen does not have d orbitals in its valence shell.
 - C nitrogen chloride covalent bonds are very weak.
 - D nitrogen has only two orbitals in its outer shell.

- 14 Which graph shows the correct trend in electronegativity down a group?



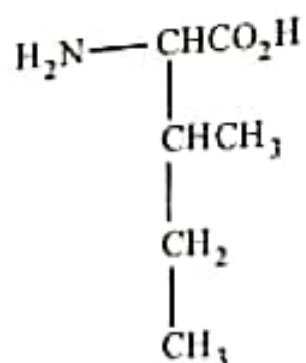
- 15 Ethene and butane are similar in that they

- A burn in oxygen to produce water and carbon dioxide.
- B are hydrogenated using a suitable catalyst.
- C polymerise under suitable conditions.
- D are obtained by the dehydration of alcohols.

- 16 Which one is a termination step in the reaction between ethane and chlorine?

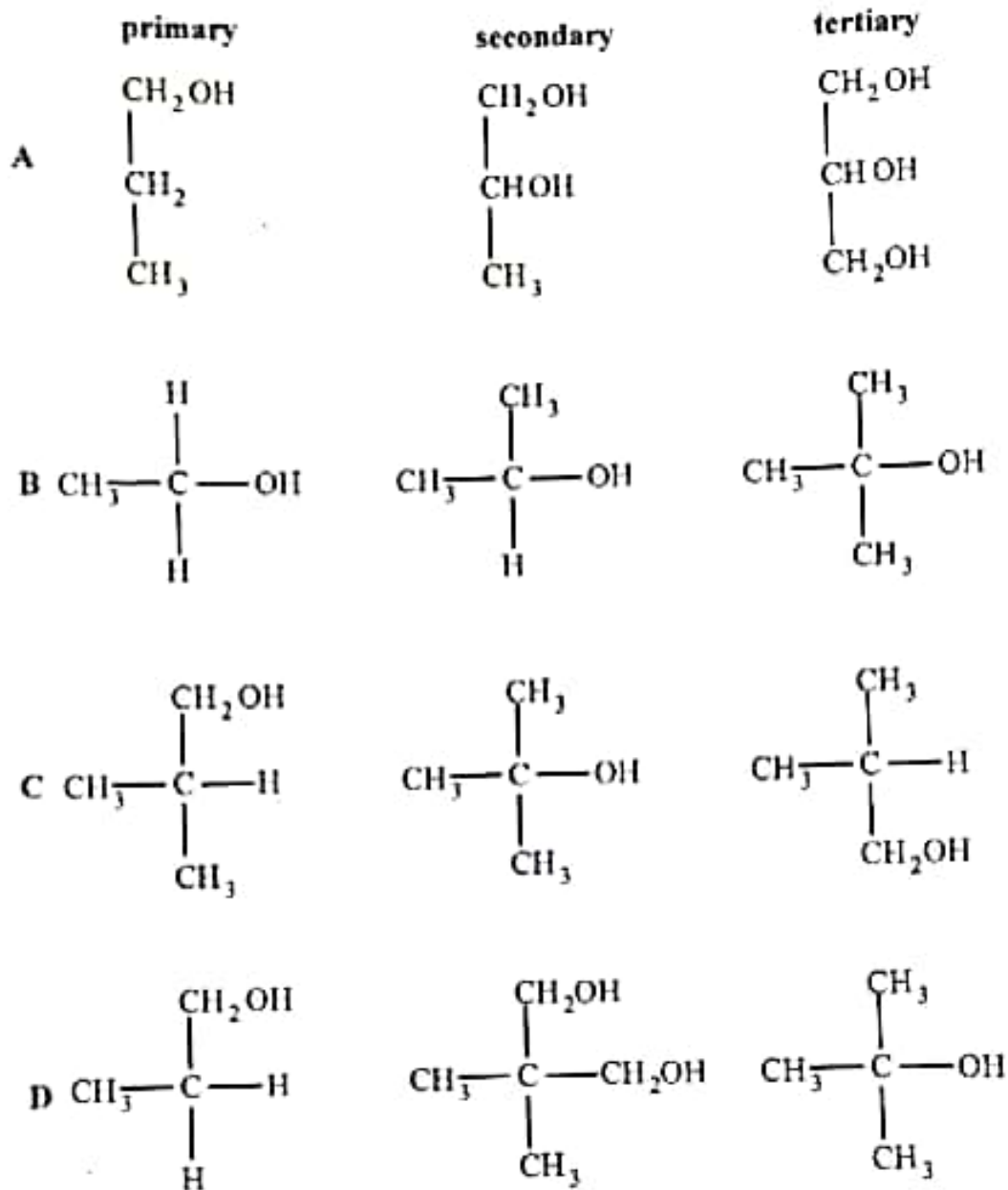
- A $\text{CH}_3\text{CH}_2\cdot + \text{Cl}\cdot \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
- B $\text{Cl}_2 \rightarrow \text{Cl}\cdot + \text{Cl}\cdot$
- C $\text{CH}_3\text{CHCl}\cdot + \text{Cl}_2 \rightarrow \text{CH}_3\text{CHCl}_2 + \text{Cl}\cdot$
- D $\text{CH}_3\text{CH}_2\cdot + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{Cl}\cdot$

- 17 What type of isomerism is shown by the organic compound of the structure

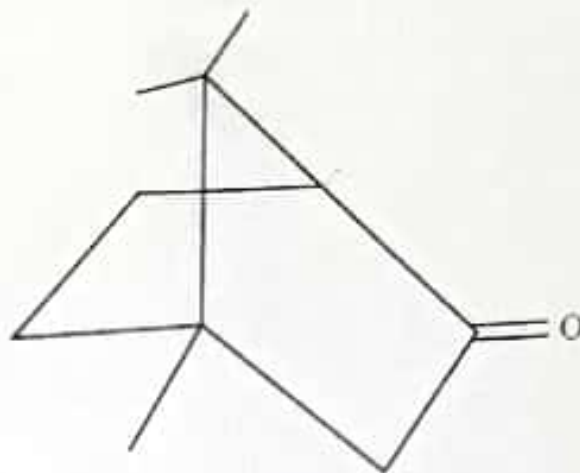


- A geometric
B positional
C optical
D structural
- 18 Which one is **not** a product of the reaction of methane with chlorine in the presence of U.V light?
- A 1-chloropropane
B dichloromethane
C trichloromethane
D chloroethene
- 19 Why is phenylamine a weaker base than ethylamine in water?
- A Phenylamine is less soluble in water.
B Phenylamine does not form hydrogen bonds with water.
C In phenylamine, the protons are attracted to the benzene ring rather than to the nitrogen atom.
D In phenylamine, the lone pair of electrons on the nitrogen atom is delocalised.
- 20 The mechanism involved when chlorine reacts with propene in an organic solvent is
- A electrophilic addition.
B electrophilic substitution
C nucleophilic addition
D nucleophilic substitution
- 21 Which property can be used to identify the organic compounds of molecules with formula $\text{C}_3\text{H}_9\text{N}$.
- A chirality
B solubility
C reactions with $\text{C}_2\text{H}_5\text{COOH}$.
D reactions with NaNO_2 .

Which set of alcohols correctly shows a primary, a secondary and a tertiary alcohol?



23 The skeletal structure of camphor is shown.



How many chiral centres does camphor have?

- A 0
- B 1
- C 2
- D 3

24 How many different aldehydes have the molecular formula $C_5H_{10}O$?

- A 2
- B 3
- C 4
- D 5

25 The isoelectric pH of three amino acids, X, Y and Z, are 7.50, 6.00 and 4.50 respectively.

What is the correct arrangement of the amino acids during electrophoresis when buffered at a pH of 6.00?

	(-)				(+)
A		z	x	y	
B		z	y	x	
C		x	y	z	
D		x	z	y	

- 26 Which one is an effect of water pollution by oil in seas?
- A increase in the sea level
 - B reduction in marine species
 - C reduction in global warming
 - D increase in eutrophication
- 27 Which pollutant is formed in the internal combustion engine and leads to the formation of acid rain?
- A Carbon monoxide
 - B Carbon
 - C Nitrogen monoxide
 - D Sulphur dioxide
- 28 Nano materials **cannot** exist in the form of
- A suspensions.
 - B liquids.
 - C colloids.
 - D solids.
- 29 Which one is **not** a Platinum Group Metal (PGM)?
- A osmium
 - B ruthenium
 - C rhodium
 - D cadmium
- 30 The $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ ion is blue whereas $[\text{CuCl}_4]^{2-}$ ion is yellow.
- Which statement explains the difference in colour?
- A difference in the number of electrons in copper
 - B difference in shapes of the complex ions
 - C difference in the energy gaps between the d-orbitals
 - D difference in the charge of the complex ions

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which molecule(s) is/are linear in shape?

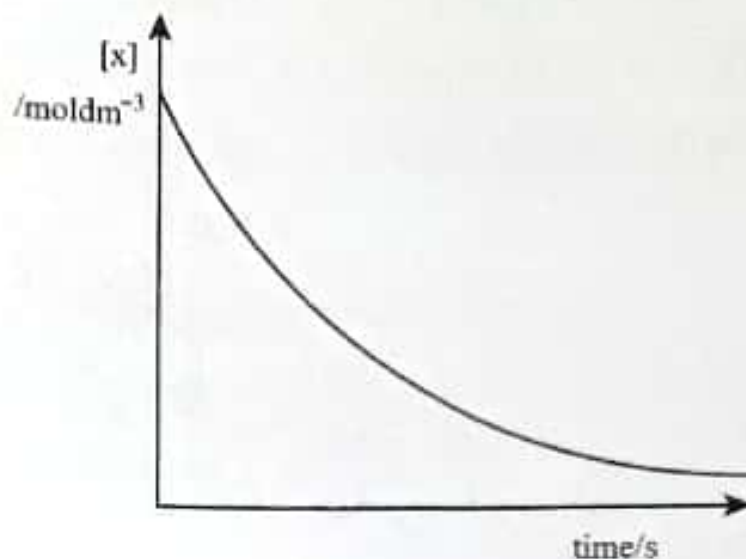
1. BeCl_2
2. CO_2
3. Cl_2O

32

The reaction shown was found to be first order:



The graph shows how the $[X]$ varied with time.



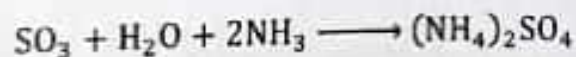
The graph can be used to determine the

1. rate of reaction at any given time.
2. order of reaction.
3. half-life of the reaction.

33 Which metal(s) is/are deposited at the cathode when 290 C of electricity are passed through 1×10^{-3} moles of a molten compound?

1. aluminium
2. chromium
3. sodium

34 Ammonium sulphate can be prepared by the reaction:



What occurs during this reaction

1. acid/base neutralisation.
2. dative bond formation.
3. redox reaction.

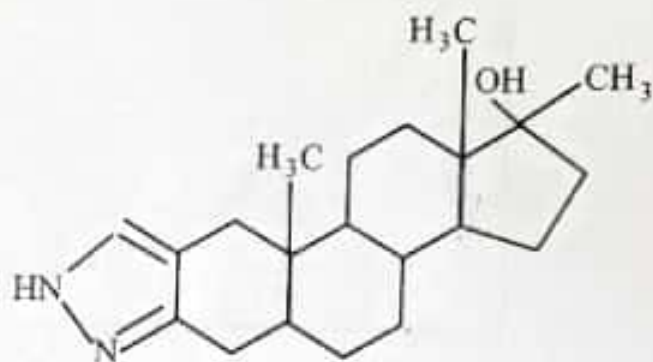
6031/1 N2019

. [Turn over

35 What happens when calcium nitrate decomposes?

1. a colourless gas is observed
2. a gas that relights a glowing splint is produced
3. the mass of calcium nitrate decreases

36 The structure of stanozolol is shown.



stanozolol

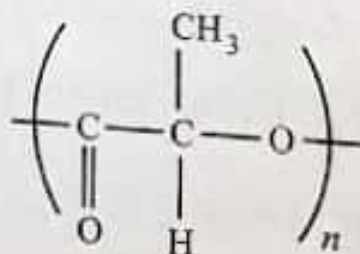
Which statement(s) about stanozolol is/are correct?

1. It has 2 chiral centres.
2. It is a tertiary alcohol.
3. It decolourises KMnO_4 .

37 The molecular formula of lactic acid is $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$.

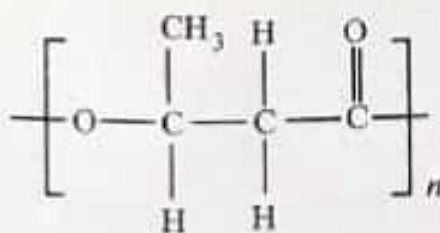
Which statement is/are true about lactic acid?

1. It can be oxidised to a lactone
2. It can polymerise to form



3. It shows geometrical isomerism

- 38 The polymer shown occurs in bacteria as cell storage material.



Which deduction(s) about this substance can be made from the structure?

1. It is a polyester.
2. It could be readily made from $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$.
3. It reacts with phosphorus pentachloride to give fumes of hydrogen chloride.

- 39 The outer electronic structure of an element is $3d^5 4s^2$.

The element

1. forms coloured ions.
2. forms complex ions.
3. has a low melting point.

- 40 The benefit(s) of reusing waste is/are

1. reduces costs of waste disposal
2. reduces environmental damage
3. saves energy